

Curved Spin(4,4) Bundle, Vielbein, and Canonical Spin Connection

A standalone exact construction on a curved split-signature eight-manifold

Reproducible exact-real implementation

September 2026

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Abstract

This document defines one real rank-16 spinor bundle over an explicit curved 8-dimensional pseudo-Riemannian base of signature $(4,4)$. It distinguishes coordinate indices from orthonormal tangent-space indices, distinguishes the curved metric $\mathbf{g}$ from the constant tangent metric $\mathbf{eta}$, constructs a diagonal vielbein, derives the Levi-Civita Christoffel symbols and spin connection from the vielbein postulate, defines eight exact real 16 by 16 gamma matrices, and constructs the spinor covariant derivative. Every formula is evaluated for the chosen curved metric. Independent Wolfram and Python implementations verify the complete vielbein postulate, connection antisymmetry, the spinor bilinear, chirality preservation, and the contracted homogeneous Dirac operator.

The construction is mathematical. A metric with four positive and four negative directions is not standard one-time physical spacetime. The chosen coordinate $\mathbf{t}$ is one timelike evolution coordinate, while each constant- $\mathbf{t}$ slice still has signature $(4,3)$. No claim of observational viability, causal stability, or uniqueness is made.

1 Geometric data and index conventions

Let $\mathbf{M}$ be an oriented, time-oriented, smooth 8-manifold equipped with a pseudo-Riemannian metric $\mathbf{g}$ of signature $(4,4)$. Assume that $\mathbf{M}$ admits a spin structure. Equivalently for the present purpose, the oriented pseudo-orthonormal frame bundle has a lift through the double covering

$$\mathrm{Spin}(4,4) \longrightarrow \mathrm{SO}_0(4,4).$$

Greek indices $\mathbf{mu}, \mathbf{nu}, \mathbf{rho}, \dots$ label coordinate components on $\mathbf{M}$. Latin indices $\mathbf{a}, \mathbf{b}, \mathbf{c}, \dots$ label components in a local pseudo-orthonormal tangent frame. Both range from 0 through 7. The ordered coordinates and tangent labels are

$$(\mathbf{x0}, \mathbf{x1}, \mathbf{x2}, \mathbf{x3}, \mathbf{t}, \mathbf{y1}, \mathbf{y2}, \mathbf{y3}).$$

The zero-based index of $\mathbf{t}$ is 4. The remaining seven indices are

$$\mathbf{I} = \{0, 1, 2, 3, 5, 6, 7\}.$$

The constant tangent-space metric is

$$\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1).$$

It is flat and independent of position. Calling it a $(4,4)$ Minkowski metric is common in this project, although "split pseudo-Euclidean metric" is less likely to be confused with the usual $(3,1)$ Minkowski metric.

The curved coordinate metric is a field

$$g = g_{\mu\nu}(x) dx^\mu \otimes dx^\nu.$$

Its entries can vary from point to point and its Levi-Civita curvature need not vanish. The two metrics do not live on competing spaces: $\mathbf{eta}$ gives the matrix of $\mathbf{g}$ in an orthonormal tangent frame, while $\mathbf{g}_{\mu\nu}$ gives the same bilinear form in a coordinate basis.

2 The spinor fiber bundle

Let

$$\pi_{SO} : P_{SO_0(4,4)}(M) \longrightarrow M$$

be the principal bundle of oriented pseudo-orthonormal frames. A spin structure is a principal bundle

$$\pi_{\text{Spin}} : P_{\text{Spin}(4,4)}(M) \longrightarrow M$$

together with a fiberwise two-to-one map to the frame bundle that intertwines the right group actions.

Let $\Delta_{\mathbb{R}}$ be the real 16-dimensional irreducible module of the full Clifford algebra $\mathbf{Cl}(4,4)$. Restriction to the even Clifford algebra and to $\mathbf{Spin}(4,4)$ decomposes it as

$$\Delta_{\mathbb{R}} = \Delta_+ \oplus \Delta_-, \quad \dim_{\mathbb{R}} \Delta_+ = \dim_{\mathbb{R}} \Delta_- = 8.$$

The requested fiber bundle is the associated real spinor bundle

$$\boxed{\mathcal{S} = P_{\text{Spin}(4,4)}(M) \times_{\rho} \Delta_{\mathbb{R}} \longrightarrow M.}$$

A spinor field is a smooth section

$$\psi \in \Gamma(\mathcal{S}).$$

The chirality operator splits the bundle into two rank-8 subbundles,

$$\mathcal{S} = \mathcal{S}_+ \oplus \mathcal{S}_-,$$

so locally

$$\psi = \psi_+ \oplus \psi_-.$$

These are the two inequivalent real half-spin fields called type 1 and type 2 in the task statement. They are not two unrelated bundles in this construction; they are the two chiral summands of one rank-16 real spinor bundle. The spin connection is even and therefore preserves each summand. Odd Clifford multiplication exchanges them.

A global spinor bundle requires the spin-structure assumption. A local vielbein always exists on a sufficiently small coordinate neighborhood, but a single global vielbein need not exist on an arbitrary spin manifold.

3 Vielbein terminology and the metric relation

A vielbein is a local pseudo-orthonormal frame field. The word "vierbein" literally refers to four dimensions. In eight dimensions the specific word is "achtbein"; "vielbein" is the standard dimension-independent term. It is therefore imprecise to call an 8-dimensional frame an "8-dimensional vierbein."

Write the coframe as

$$e^a = e_\mu{}^a dx^\mu$$

and its inverse as

$$e_a = e_a{}^\mu \partial_\mu.$$

They obey

$$e_a{}^\mu e_\mu{}^b = \delta_a{}^b, \quad e_\mu{}^a e_a{}^\nu = \delta_\mu{}^\nu.$$

The defining metric relation is

$$\boxed{g_{\mu\nu} = e_\mu{}^a \eta_{ab} e_\nu{}^b.}$$

In matrix notation this is

$$g = e \eta e^\top.$$

The vielbein is not unique. If $\Lambda(x)$ is a local $SO(4,4)$ transformation, then

$$e'_\mu{}^a = e_\mu{}^b \Lambda_b{}^a$$

gives the same metric. Thus a "canonical frame" means a frame selected by an explicit gauge convention for a specified metric, not a frame uniquely forced by g on every manifold.

4 The explicit curved (4,4) metric

To make every requested component computable, choose a positive smooth scale factor $a(t)$ and the cohomogeneity-one metric

$$ds^2 = a(t)^2 \left[(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2 - (dy^1)^2 - (dy^2)^2 - (dy^3)^2 \right] - dt^2.$$

In the ordered coordinates,

$$g_{\mu\nu} = \text{diag} \left(a^2, a^2, a^2, a^2, -1, -a^2, -a^2, -a^2 \right).$$

There are four positive and four negative entries. If $\mathbf{a}$ is nonconstant, the metric is curved. Constant- $\mathbf{t}$ hypersurfaces have signature $(4, 3)$, so they are not ordinary Riemannian spatial slices.

Define

$$H(t) = \frac{\dot{a}(t)}{a(t)}.$$

This resembles the Hubble parameter of a flat FLRW ansatz, but here it is the expansion rate of a seven-dimensional split-signature transverse metric.

5 Step 1: the selected canonical frame field

Choose the diagonal coframe gauge

$$e_\mu^a = \text{diag}(a, a, a, a, 1, a, a, a).$$

Its inverse is

$$e_a^\mu = \text{diag}(a^{-1}, a^{-1}, a^{-1}, a^{-1}, 1, a^{-1}, a^{-1}, a^{-1}).$$

Direct multiplication gives

$$\eta e e^\top = \text{diag}(a^2, a^2, a^2, a^2, -1, -a^2, -a^2, -a^2) = g.$$

In differential-form notation,

$$e^4 = dt, \quad e^i = a(t) dx^i \quad (i \in I),$$

where the three $\mathbf{y}$ coordinates are included in the symbol $\mathbf{dx}^{\mathbf{i}}$ according to the fixed ordered index set.

This frame is selected because it is diagonal, orientation-compatible, and reduces to the identity frame when $\mathbf{a}=1$. Local $\text{SO}(4, 4)$ transforms produce gauge-equivalent frames and transformed spin-connection components.

6 Levi-Civita Christoffel symbols

The torsion-free metric-compatible coordinate connection is

$$\Gamma^\rho{}_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}).$$

Only time derivatives are nonzero. For each transverse index i in I , the complete nonzero inventory is

$$\boxed{\Gamma^4{}_{ii} = \eta_{ii}a\dot{a}, \quad \Gamma^i{}_{4i} = \Gamma^i{}_{i4} = H.}$$

There are $7*3=21$ nonzero coordinate components. The sign in $\Gamma^4{}_{ii}$ follows the tangent signature:

- for $i=0,1,2,3$, $\Gamma^4{}_{ii}=+a \dot{a}$;
- for $i=5,6,7$, $\Gamma^4{}_{ii}=-a \dot{a}$.

No summation over i is intended in these displayed component formulas.

7 Step 2: canonical Levi-Civita spin connection

The vielbein postulate is

$$\partial_\mu e_\nu{}^a - \Gamma^\rho{}_{\mu\nu} e_\rho{}^a + \omega_\mu{}^a{}_b e_\nu{}^b = 0.$$

Multiplying by the inverse frame solves it directly:

$$\boxed{\omega_\mu{}^a{}_b = e_b{}^\nu (\Gamma^\rho{}_{\mu\nu} e_\rho{}^a - \partial_\mu e_\nu{}^a).}$$

Lower the first tangent index with η_a :

$$\omega_{\mu ab} = \eta_{ac} \omega_\mu{}^c{}_b.$$

Metric compatibility gives

$$\omega_{\mu ab} = -\omega_{\mu ba}.$$

For the chosen frame, all components with $\mu=4$ vanish. For every transverse i , the complete nonzero lowered components are

$$\boxed{\omega_{ii4} = \eta_{ii}\dot{a}, \quad \omega_{i4i} = -\eta_{ii}\dot{a}.}$$

There are 7*2=14 nonzero lowered components. Explicitly,

```
omega_0,04 = +adot    omega_0,40 = -adot
omega_1,14 = +adot    omega_1,41 = -adot
omega_2,24 = +adot    omega_2,42 = -adot
omega_3,34 = +adot    omega_3,43 = -adot
omega_5,54 = -adot    omega_5,45 = +adot
omega_6,64 = -adot    omega_6,46 = +adot
omega_7,74 = -adot    omega_7,47 = +adot
```

Here a comma separates the coordinate index from the two tangent indices. Substitution into every one of the 8*8*8=512 vielbein-postulate components returns zero exactly.

The task's Step 4 repeats the Step 2 equation. No new connection is required: the Levi-Civita construction is dimension-independent, and the formula above is already its direct 8-dimensional generalization. Introducing a second "canonical" connection without extra torsion or nonmetricity data would be mathematically unjustified.

8 Exact real 16 by 16 gamma matrices

Define

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy

$$P^2 = I_2, \quad N^2 = -I_2, \quad G^2 = I_2,$$

and anticommute pairwise. On the fourth tensor power of $\mathbb{R}^2$, define

$$\gamma_k^+ = G^{\otimes(k-1)} \otimes P \otimes I_2^{\otimes(4-k)},$$

$$\gamma_k^- = G^{\otimes(k-1)} \otimes N \otimes I_2^{\otimes(4-k)}, \quad k = 1, 2, 3, 4.$$

The ordered list is

$$(\gamma_1^+, \gamma_2^+, \gamma_3^+, \gamma_4^+, \gamma_1^-, \gamma_2^-, \gamma_3^-, \gamma_4^-).$$

In zero-based indexing, the evolution-time generator is $\mathbf{gamma}^4$, the first negative generator and fifth matrix in this list. The matrices obey

$$\gamma^a \gamma^b + \gamma^b \gamma^a = 2\eta^{ab} I_{16}.$$

Curved gamma matrices are

$$\gamma^\mu(x) = e_a{}^\mu(x)\gamma^a,$$

and satisfy

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}I_{16}.$$

9 Step 3: spinor covariant derivative

Define

$$\Omega_\mu = \frac{1}{8}\omega_{\mu ab}[\gamma^a, \gamma^b]$$

and

$$\boxed{D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi.}$$

The factor 1/8 is correct when the repeated indices **a, b** are summed over all ordered pairs. Since **omega_muab** and the commutator are both antisymmetric, the (**a, b**) and (**b, a**) terms are equal. If one instead writes only a sum over **a < b**, the equivalent coefficient is 1/4:

$$\frac{1}{8}\sum_{a,b}\omega_{\mu ab}[\gamma^a, \gamma^b] = \frac{1}{4}\sum_{a<b}\omega_{\mu ab}[\gamma^a, \gamma^b].$$

For this frame,

$$\Omega_4 = 0,$$

and for each transverse index,

$$\boxed{\Omega_i = \frac{1}{2}\eta_{ii}\dot{a}\gamma^i\gamma^4.}$$

The chirality operator is

$$\omega_{\text{chir}} = \gamma^0\gamma^1\cdots\gamma^7.$$

Every **Omega_mu** is an even Clifford element, so

$$[\Omega_\mu, \omega_{\text{chir}}] = 0.$$

The connection therefore preserves the two half-spin subbundles even though odd Clifford multiplication exchanges them.

10 Homogeneous Dirac operator

For a homogeneous spinor $\psi(t)$, transverse partial derivatives vanish but transverse spin-connection terms do not. Contracting with the curved gamma matrices gives

$$\gamma^\mu \Omega_\mu = \sum_{i \in I} \frac{1}{a} \gamma^i \left(\frac{1}{2} \eta_{ii} \dot{a} \gamma^i \gamma^4 \right).$$

Since $(\gamma^i)^2 = \eta_{ii} \mathbb{I}$, each summand equals $H \gamma^4/2$. There are seven transverse directions, hence

$$\gamma^\mu D_\mu \psi = \gamma^4 \left(\partial_t + \frac{7}{2} H \right) \psi.$$

The factor $7/2$ is the 8-dimensional analogue of the familiar $3H/2$ term in a four-dimensional spatially flat homogeneous spinor equation.

11 Real invariant spinor bilinear

The exact gamma representation admits the symmetric matrix

$$C = \gamma_1^+ \gamma_2^+ \gamma_3^+ \gamma_4^+.$$

It obeys

$$C^\top = C, \quad C^2 = I_{16}, \quad (\gamma^a)^\top C = -C \gamma^a.$$

Its eigenvalue signature is $(8,8)$, so it is nondegenerate but not positive definite. Define

$$\bar{\psi} = \psi^\top C, \quad S = \bar{\psi} \psi = \psi^\top C \psi.$$

For a local spin transformation R generated by bivectors,

$$R^\top C R = C,$$

so S is locally $\text{Spin}(4,4)$ invariant. This is the covariant bilinear used by the coupled gravitational model. It must not be confused with the positive Euclidean quantity $\psi^\top \psi$ used in a previous background-only numerical reconstruction.

Because $\mathbf{C}$ is symmetric, a single Grassmann-odd spinor would satisfy $\mathbf{\psi}^T \mathbf{C} \mathbf{\psi} = 0$. The nonlinear model that uses nonzero $\mathbf{S}$ therefore treats $\mathbf{\psi}$ as a commuting classical real spinor field, an effective classical order parameter, rather than a quantized fermionic field.

12 Exact verification strategy

Two implementations derive the geometry independently.

The Python generator differentiates the metric and frame using exact rational arithmetic at the nontrivial sample

$$a = \frac{3}{2}, \quad \dot{a} = \frac{2}{5}, \quad H = \frac{4}{15}.$$

It checks all 512 vielbein-postulate components, all connection antisymmetry components, the exact invariant bilinear, the chirality commutators, and the full 16 by 16 slash-connection identity.

The Wolfram implementation performs symbolic differentiation with an unspecified function $\mathbf{a}(\mathbf{t})$. It independently derives the metric, Christoffels, spin connection, and spin matrices and checks the same symbolic identities without reading the Python result.

The independent Python checker reads both reports, reconstructs the Christoffels and spin connection again from metric/frame derivatives, verifies source hashes, and compares exact half-spin indices to volume-element eigenspaces.

Expected exact totals are

```
base_dimension=8
signature=(4,4)
spinor_fiber_dimension=16
half_spin_dimensions=(8,8)
nonzero_Christoffel_components=21
nonzero_lowered_spin_connection_components=14
spinor_bilinear_signature=(8,8)
Python_generator_checks=12
Wolfram_symbolic_checks=11
independent_checker_checks=16
```

13 Complete Windows commands

Open PowerShell and run every command below from a clean checkout:

```
Set-Location C:\Users\nsh\Developer\code\vscode\dirac
git submodule update --init --recursive

python scripts\backup_files.py '
    artifacts\curved-spin-geometry\geometry.json '
    artifacts\curved-spin-geometry\wolfram-report.json '
```

```

provenance\CURVED_SPIN_BUNDLE.tex '
provenance\CURVED_SPIN_BUNDLE.pdf

New-Item -ItemType Directory -Force -Path '
    build\curved-spin-geometry\pdf-a, '
    build\curved-spin-geometry\pdf-b | Out-Null

python scripts\build_curved_spin_geometry.py
python scripts\build_curved_spin_geometry.py '
    --output build\curved-spin-geometry\geometry-repeat.json

wolframscript -file scripts\verify_curved_spin_geometry.wls -- '
    artifacts\curved-spin-geometry\wolfram-report.json
wolframscript -file scripts\verify_curved_spin_geometry.wls -- '
    build\curved-spin-geometry\wolfram-report-repeat.json

python scripts\check_curved_spin_geometry.py
python -m unittest discover -s tests -p test_curved_spin_geometry.py -v
python -m unittest discover -s tests -v

python scripts\build_dissertation_tex.py --strip-heading-numbers '
    --input provenance\CURVED_SPIN_BUNDLE.md '
    --output provenance\CURVED_SPIN_BUNDLE.tex
python scripts\build_dissertation_tex.py --strip-heading-numbers '
    --input provenance\CURVED_SPIN_BUNDLE.md '
    --output build\curved-spin-geometry\CURVED_SPIN_BUNDLE-repeat.tex

pdflatex -interaction=nonstopmode -halt-on-error '
    -jobname=CURVED_SPIN_BUNDLE '
    -output-directory=build\curved-spin-geometry\pdf-a '
    provenance\CURVED_SPIN_BUNDLE.tex
pdflatex -interaction=nonstopmode -halt-on-error '
    -jobname=CURVED_SPIN_BUNDLE '
    -output-directory=build\curved-spin-geometry\pdf-a '
    provenance\CURVED_SPIN_BUNDLE.tex
pdflatex -interaction=nonstopmode -halt-on-error '
    -jobname=CURVED_SPIN_BUNDLE '
    -output-directory=build\curved-spin-geometry\pdf-a '
    provenance\CURVED_SPIN_BUNDLE.tex

pdflatex -interaction=nonstopmode -halt-on-error '
    -jobname=CURVED_SPIN_BUNDLE '
    -output-directory=build\curved-spin-geometry\pdf-b '
    build\curved-spin-geometry\CURVED_SPIN_BUNDLE-repeat.tex
pdflatex -interaction=nonstopmode -halt-on-error '
    -jobname=CURVED_SPIN_BUNDLE '
    -output-directory=build\curved-spin-geometry\pdf-b '
    build\curved-spin-geometry\CURVED_SPIN_BUNDLE-repeat.tex
pdflatex -interaction=nonstopmode -halt-on-error '
    -jobname=CURVED_SPIN_BUNDLE '
    -output-directory=build\curved-spin-geometry\pdf-b '
    build\curved-spin-geometry\CURVED_SPIN_BUNDLE-repeat.tex

python scripts\check_provenance_pdf.py --edition curved-spin-bundle '

```

```

build\curved-spin-geometry\pdf-a\CURVED_SPIN_BUNDLE.pdf '
--repeat build\curved-spin-geometry\pdf-b\CURVED_SPIN_BUNDLE.pdf

$texA = (Get-FileHash '
  provenance\CURVED_SPIN_BUNDLE.tex '
  -Algorithm SHA256).Hash
$texB = (Get-FileHash '
  build\curved-spin-geometry\CURVED_SPIN_BUNDLE-repeat.tex '
  -Algorithm SHA256).Hash
if ($texA -ne $texB) { throw "Curved-spin LaTeX changed bytes" }
$pdfA = (Get-FileHash '
  build\curved-spin-geometry\pdf-a\CURVED_SPIN_BUNDLE.pdf '
  -Algorithm SHA256).Hash
$pdfB = (Get-FileHash '
  build\curved-spin-geometry\pdf-b\CURVED_SPIN_BUNDLE.pdf '
  -Algorithm SHA256).Hash
if ($pdfA -ne $pdfB) { throw "Curved-spin PDF changed bytes" }

$warningPattern = '^!|LaTeX Warning|Package .* Warning|'
$warningPattern += 'Overfull|Underfull|Undefined control sequence'
$issues = Select-String '
  -Path build\curved-spin-geometry\pdf-a\CURVED_SPIN_BUNDLE.log, '
    build\curved-spin-geometry\pdf-b\CURVED_SPIN_BUNDLE.log '
  -Pattern $warningPattern
if ($issues) { throw "Curved-spin LaTeX warnings remain" }

Copy-Item build\curved-spin-geometry\pdf-a\CURVED_SPIN_BUNDLE.pdf '
  provenance\CURVED_SPIN_BUNDLE.pdf -Force

.\scripts\verify_phase5_curved_spin_gravity.ps1

```

The final command runs the complete exact geometry, coupled numerical study, two provenance builds, all tests, all byte comparisons, and all warning scans. Its final line must be

```
phase5_curved_spin_gravity_verification=OK
```

14 Complete Git Bash or WSL commands

Run every command below:

```

if [[ -d /c/Users/nsh/Developer/code/vscode/dirac ]]; then
  cd /c/Users/nsh/Developer/code/vscode/dirac
else
  cd /mnt/c/Users/nsh/Developer/code/vscode/dirac
fi
git submodule update --init --recursive
source ./scripts/resolve_wolframscrip.sh
python_command="$(resolve_windows_command python.exe)"
pdflatex_command="$(resolve_miktex_pdflatex)"

```

```

"$python_command" scripts/backup_files.py \
  artifacts/curved-spin-geometry/geometry.json \
  artifacts/curved-spin-geometry/wolfram-report.json \
  provenance/CURVED_SPIN_BUNDLE.tex \
  provenance/CURVED_SPIN_BUNDLE.pdf

rm -rf build/curved-spin-geometry/pdf-a build/curved-spin-geometry/pdf-b
mkdir -p build/curved-spin-geometry/pdf-a build/curved-spin-geometry/pdf-b

"$python_command" scripts/build_curved_spin_geometry.py
"$python_command" scripts/build_curved_spin_geometry.py \
  --output build/curved-spin-geometry/geometry-repeat.json

wolframscript_command="$ (resolve_wolframscript)"
"$wolframscript_command" -file scripts/verify_curved_spin_geometry.wls -- \
  artifacts/curved-spin-geometry/wolfram-report.json
"$wolframscript_command" -file scripts/verify_curved_spin_geometry.wls -- \
  build/curved-spin-geometry/wolfram-report-repeat.json

"$python_command" scripts/check_curved_spin_geometry.py
"$python_command" -m unittest discover -s tests \
  -p test_curved_spin_geometry.py -v
"$python_command" -m unittest discover -s tests -v

"$python_command" scripts/build_dissertation_tex.py \
  --strip-heading-numbers \
  --input provenance/CURVED_SPIN_BUNDLE.md \
  --output provenance/CURVED_SPIN_BUNDLE.tex
"$python_command" scripts/build_dissertation_tex.py \
  --strip-heading-numbers \
  --input provenance/CURVED_SPIN_BUNDLE.md \
  --output build/curved-spin-geometry/CURVED_SPIN_BUNDLE-repeat.tex

for pass in 1 2 3; do
  "$pdflatex_command" -interaction=nonstopmode -halt-on-error \
    -jobname=CURVED_SPIN_BUNDLE \
    -output-directory=build/curved-spin-geometry/pdf-a \
    provenance/CURVED_SPIN_BUNDLE.tex
  "$pdflatex_command" -interaction=nonstopmode -halt-on-error \
    -jobname=CURVED_SPIN_BUNDLE \
    -output-directory=build/curved-spin-geometry/pdf-b \
    build/curved-spin-geometry/CURVED_SPIN_BUNDLE-repeat.tex
done

"$python_command" scripts/check_provenance_pdf.py \
  --edition curved-spin-bundle \
  build/curved-spin-geometry/pdf-a/CURVED_SPIN_BUNDLE.pdf \
  --repeat build/curved-spin-geometry/pdf-b/CURVED_SPIN_BUNDLE.pdf

tex_a="$ (sha256sum provenance/CURVED_SPIN_BUNDLE.tex | cut -d' ' -f1)"
tex_b="$ (sha256sum \
  build/curved-spin-geometry/CURVED_SPIN_BUNDLE-repeat.tex | \
  cut -d' ' -f1)"
test "$tex_a" = "$tex_b"

```

```
pdf_a="$(sha256sum \
  build/curved-spin-geometry/pdf-a/CURVED_SPIN_BUNDLE.pdf | \
  cut -d' ' -f1)"
pdf_b="$(sha256sum \
  build/curved-spin-geometry/pdf-b/CURVED_SPIN_BUNDLE.pdf | \
  cut -d' ' -f1)"
test "$pdf_a" = "$pdf_b"
warning_pattern='^!|LaTeX Warning|Package .* Warning|'
warning_pattern+= 'Overfull|Underfull|Undefined control sequence'
! grep -Ei "$warning_pattern" \
  build/curved-spin-geometry/pdf-a/CURVED_SPIN_BUNDLE.log \
  build/curved-spin-geometry/pdf-b/CURVED_SPIN_BUNDLE.log
cp build/curved-spin-geometry/pdf-a/CURVED_SPIN_BUNDLE.pdf \
  provenance/CURVED_SPIN_BUNDLE.pdf

bash ./scripts/verify_phase5_curved_spin_gravity.sh
```

15 Files created and their roles

```
wolfram/CurvedSpinGeometry.wl
scripts/verify_curved_spin_geometry.wls
scripts/build_curved_spin_geometry.py
scripts/check_curved_spin_geometry.py
artifacts/curved-spin-geometry/geometry.json
artifacts/curved-spin-geometry/wolfram-report.json
provenance/CURVED_SPIN_BUNDLE.md
provenance/CURVED_SPIN_BUNDLE.tex
provenance/CURVED_SPIN_BUNDLE.pdf
```

The Wolfram and Python definitions are independent implementations of the same mathematical equations. The JSON files are machine-readable exact evidence. The Markdown, LaTeX, and PDF contain the complete standalone human explanation and commands.

16 Limitations

The frame is canonical only relative to the chosen diagonal gauge and metric ansatz. It is not a unique frame for every $(4,4)$ metric. The base is assumed to possess the required orientation, time orientation, and spin structure. The explicit chart and frame describe the chosen coordinate domain; global topology is not classified.

The constant- τ slices are split signature $(4,3)$. The construction is a valid pseudo-Riemannian and spin-geometric model, but it is not ordinary Lorentzian cosmology. Any physical interpretation requires separate analyses of causality, hyperbolicity, ghosts, boundary conditions, compactification, perturbations, and observations.

17 Conclusion

One precise fiber bundle has been defined:

$$\mathcal{S} = P_{\text{Spin}(4,4)}(M) \times_{\rho} (\Delta_+ \oplus \Delta_-).$$

The flat tangent metric $\mathbf{eta}$, curved metric $\mathbf{g}$, and selected vielbein $\mathbf{e}$ are related by $\mathbf{g} = \mathbf{e} \mathbf{eta} \mathbf{e}^T$. The Levi-Civita spin connection follows uniquely from the stated frame gauge and vielbein postulate. The requested covariant derivative with coefficient $1/8$ is correct under an all-ordered-pairs tangent index sum. Exact Python and Wolfram calculations agree on every nonzero connection component and on the homogeneous $7H/2$ spin-connection term.